

- 2 Fig. 2.1 is a scanning electron micrograph showing a macrophage engulfing some bacteria.

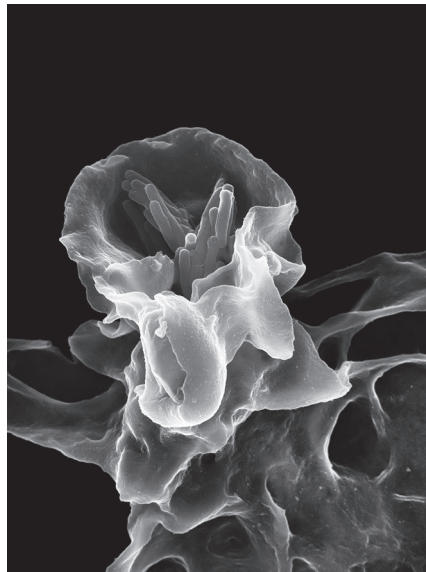


Fig. 2.1

- (a) (i) Describe how macrophages engulf bacteria.

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- (ii) Explain the functions of lysosomes in cells such as macrophages.

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Question 2 continues on page 6.

- (b) Fig. 2.2 shows the reported number of new cases of tuberculosis (TB) in the USA and the number of new cases per 100 000 of the population of the USA between 1993 and 2018.

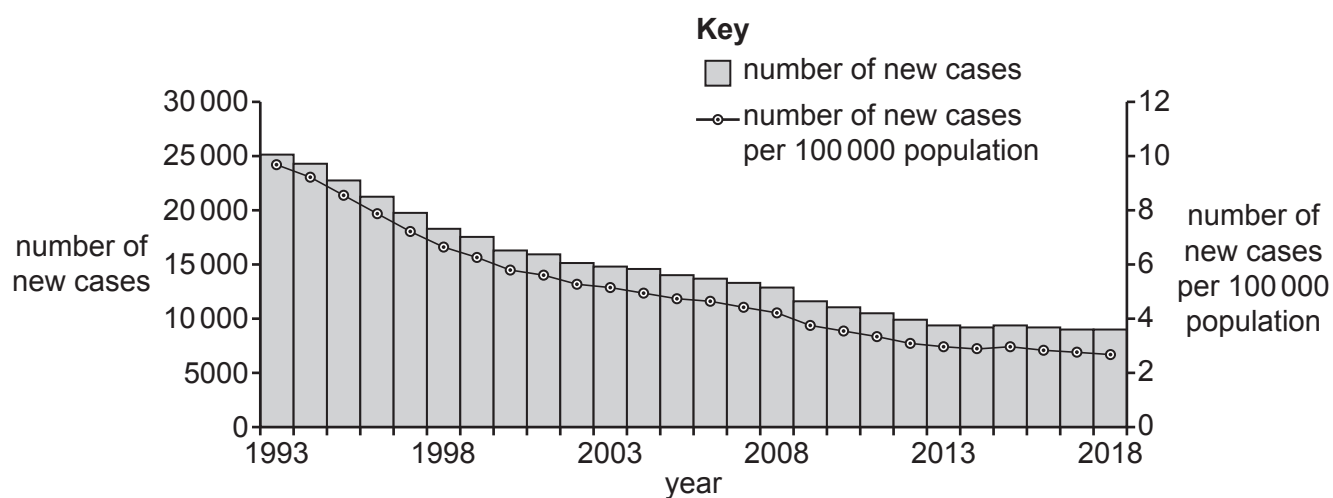


Fig. 2.2

- (i) Calculate the percentage change in the number of new cases of TB in the USA between 1993 and 2018.

Show all your working.

answer % [3]

- (ii) Use Fig. 2.2 to describe the trend in the number of new cases of TB in the USA between 1993 and 2018.

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- (c) Suggest an advantage of calculating the number of new cases per 100 000 each year in the prevention and control of TB across the world.

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- (d) TB is endemic (always present) in many populations across the world and many countries have high numbers of cases.

State **two** reasons why it is difficult to reduce the number of cases of TB across the world.

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[Total: 14]